

Received XX Month, XXXX; revised XX Month, XXXX; accepted XX Month, XXXX; Date of publication XX Month, XXXX; date of current version XX Month, XXXX.

Digital Object Identifier 10.1109/XXXX.2022.1234567

Cyber-Resilient Perception: Safeguarding Autonomous Vehicles with Trust-Aware Sensor Fusion

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ABSTRACT This study introduces a multilayered sensor fusion framework that features dynamic trust modeling to secure autonomous vehicle (AV) perception against advanced cyber-physical attacks. With increased attacks on the horizon, the current approaches lacks the ability to detect coordinated multisensor attacks, identify compromised sensors in real-time, and dynamically adjust trust levels and fusion logic accordingly. By using cross-validation, anomaly detection, and multisensor correlative consistency checking, the proposed system swiftly detects compromised sensors when incorporated with Dirichlet trust distributions, which are calculated at certain time intervals. This enables the system to dynamically minimize compromised sensors while effectively combining and fusing reliable inputs to create an accurate perception model. The model's resilience is validated by comprehensive simulation, which maintains reliable perception, while cyber-attacks, such as spoofing and jamming, actively compromise 30% of the sensors. Moreover, conducting vehicle testing under real-world conditions provides additional evidence of the effectiveness, fault tolerance, and computing efficiency in practical settings. AV safety is improved by this innovative dynamic trust modeling methodology, which guarantees a reliable perception in the event of a sensor compromise due to malicious activity. Through continuous sensor integrity monitoring, the resilient sensor fusion approach establishes a foundation for autonomous driving capabilities that are both verifiable and secure. Rapid detection and mitigation of multi-sensor attacks is a substantial improvement in the safe deployment of AVs. This work represents a step in ensuring the trustworthiness and resilience of AV systems, paving the way for safer and more secure self-driving technologies in the face of emerging cyber-physical threats.

INDEX TERMS Sensor, Reputation System, Autonomous Vehicle (AV), Dirichlet Distribution, Trust System.

I. INTRODUCTION

AUTONOMOUS vehicles (AVs) constitute an advanced technological advancement in modern transportation systems to enhance safety, operational efficiency, and accessibility across multiple domains [26]. By integrating advanced sensor frameworks and computational algorithms, AVs can perform real-time decision-making processes, enabling functionalities such as autonomous navigation, dynamic obstacle avoidance, and optimized traffic coordination [2], [5]. These AVs are integral components within smart transportation systems, with implications for transformative applications in sectors such as freight logistics and individual mobility solutions [21]. The widespread integration of AVs is

anticipated to mitigate human error-related traffic incidents, alleviate systemic traffic congestion, and provide inclusive mobility alternatives for populations underserved by traditional transportation infrastructure [25].

Despite their promising advantages, AVs face challenges, particularly in regard to the privacy and security of their sensor systems. These sensors, which are critical for environmental perception, are vulnerable to advanced cyber-physical attacks that target multiple components [27]. Such coordinated attacks have the potential to compromise sensor data integrity, resulting in erroneous decision-making and operational failures [24]. Traditional sensor fusion methodologies, although effective under controlled conditions, often

lack the resilience required to detect and mitigate these sophisticated threats, thereby exposing AVs to risks that could lead to severe consequences [12]. Addressing these vulnerabilities is essential for ensuring the safe and reliable operation of AVs in real-world environments.

The efficiency of AVs is critically reliant on multisensor systems that integrate data from various sources, including LiDAR, cameras, radar, and ultrasonic sensors. These systems provide perception of the environment, facilitating accurate object detection, distance assessment, and attack prediction. The fusion of these diverse sensor data enables AVs to construct a comprehensive understanding of their surroundings, thereby supporting informed decision-making and safe navigation across varied and dynamic conditions [23]. However, the dependency on such a wide array of sensors inherently expands the attack surface, offering adversaries multiple avenues to manipulate or disrupt the data streams. Vulnerabilities such as LiDAR spoofing, camera interference, or radar jamming pose risks to the integrity of sensor data, potentially leading to erroneous environmental mapping and operational failures [17].

Integrating trust and reputation mechanisms into sensors offers a novel approach to enhance the resilience of AVs [29]. Trust, defined as the measure of confidence in the reliability and accuracy of a system or component, is evaluated through the continuous monitoring of sensor performance patterns and historical reliability by assigning adaptive confidence scores that directly influence system responses [31]. AV systems can sustain their functionality even with deliberate adversarial manipulation or unexpected malfunctions by elevating the impact of inputs from high-confidence sensors and sidelining those that exhibit irregularities or potential tampering.

Dirichlet trust distributions are statistical methods that facilitate the dynamic adjustment of trust levels based on real-time sensor-performance assessments. This approach employs techniques such as cross-validation, anomaly detection algorithms, and correlative consistency checks to promptly identify and isolate the compromised sensors [30]. By continuously recalibrating trust values, the framework ensures that perception capabilities remain reliable and maintain accuracy even when a substantial number of sensors are targeted by cyberattacks [28].

This study aims to enhance the resilience of AV perception systems to cyber-attacks by developing a sensor fusion framework that is cognizant of trust levels and can dynamically adjust to identify and counter coordinated attacks on multiple sensors. By implementing this methodology, the AV ensures operational safety and high perception accuracy in the event of a sensor compromise. Comprehensive simulations and real-world testing were conducted to validate the framework's effectiveness, computational efficiency, and practical applicability across a wide range of scenarios, ultimately providing a solid foundation for the safe deployment

of AV systems, while also increasing their trustworthiness and reliability in the face of emerging cyber threats.

The remainder of this paper is organized as follows. Section II describes background knowledge and related work. Section III presents the system architecture and Section IV discusses the key components of the system. Finally, Section V presents conclusions and future directions.

II. BACKGROUND AND RELATED WORKS

A. Application of Autonomous Vehicle

AV networks offer numerous advantages in various areas. It can swiftly provide situational awareness [4] by collecting real-time data, identifying affected areas, and supporting search and rescue operations. This capability improves response times, enhances safety, and aids emergency personnel in decision-making.

AV networks have the potential to revolutionize the telecommunications industry by extending network coverage to remote or underserved areas. By acting as flying base stations, drones can create a wireless mesh network, enabling connectivity in areas with a limited infrastructure. In addition, it has been demonstrated that UAV-based mesh networks can be used in military environments and disaster management [8].

In precision agriculture [20], AV networks can optimize crop monitoring and management by collecting high-resolution imagery, thermal data, and sensor readings. These data can be analyzed to assess crop health, detect pest infestations, and optimize resource allocation, leading to improved yields and reduced environmental impacts.

B. Challenges in AV System

1) Battery Efficiency

Technological advancements in battery efficiency and energy harvesting techniques can address these limitations to some extent. The integration of fuel cells or battery hybrid systems is a promising way to overcome this challenge. Fuel cells, particularly hydrogen-based ones, provide higher energy density than traditional lithium-polymer batteries, enhancing flight endurance. These hybrid systems not only extend flight times but also support more energy-intensive payloads, thereby enhancing the overall mission capabilities of AVs [34].

2) Reliability and Communication

Another challenge lies in developing reliable and secure communication protocols for AV networks [3]. Ensuring data integrity, confidentiality, and resistance to cyber threats is crucial, considering the potential vulnerabilities of wireless communication in AV systems [22].

3) Scalability and Interoperability

The scalability and interoperability of AV networks need to be addressed. As the number of AVs increases, effective management of the network becomes more complex [1]. Standardized protocols, coordination mechanisms, and efficient mission-planning algorithms are essential for seamless collaboration among multiple AVs.

C. Dirichlet Distribution in Probabilistic Modeling

The Dirichlet distribution, introduced by *Johann Peter Gustav Lejeune Dirichlet*, serves as a fundamental tool in probabilistic modeling and Bayesian inference. The Dirichlet process was introduced as a distribution over other distributions, marking an advancement in nonparametric Bayesian statistics [14]. This study laid the groundwork for modern applications in probabilistic modeling.

A hierarchical Dirichlet process (HDP) was developed to extend the original concept to handle grouped data and provide a flexible framework for clustering problems [37]. This advancement has proven to be particularly valuable for handling complex data structures and uncertainty quantification. A dependent Dirichlet process was introduced to enable the modeling of distributions that evolve or space.

The mathematical properties of Dirichlet distributions, particularly their role as conjugate priors to multinomial distributions, have made them invaluable in modern machine-learning applications. In seminal work on Latent Dirichlet Allocation (LDA), Dirichlet distribution, which has become a cornerstone technique in topic modeling and data analysis, was used [7].

D. Advancement in Sensors

Recent changes in automotive sensor technology are increasing vehicle safety, autonomy, and driver experience. One innovation is the development of compact, high-resolution LiDAR (Light Detection and Ranging) systems. Companies like Luminar Technologies have introduced sensors such as the Halo, designed for seamless integration into vehicle architectures [6]. These sensors provide precise three-dimensional mapping of the vehicle's surroundings, crucial for advanced driver-assistance systems (ADAS) and autonomous driving capabilities. The collaboration between Luminar and software firms like Applied Intuition aims to streamline the testing and validation of these systems, accelerating their deployment in consumer vehicles [33].

In parallel, the emergence of hybrid optical sensors capable of simultaneous thermal and dimensional monitoring during laser processing has been highlighted [10]. These sensors employ non-contact, high-speed optical measurements to monitor physical processes, offering a non-invasive and versatile approach across various applications. The integration of such sensors into AV systems can provide real-time data on environmental conditions and vehicle dynamics, thereby enhancing the overall safety and efficiency of autonomous navigation [10].

E. Sensor Fusion Technologies and Applications

The evolution of multisensor data fusion represents the confluence of theoretical foundations and practical innovations in the field of sensor integration. The fundamental principles of multisensor integration were established by introducing a probabilistic framework that addresses environmental representation, uncertainty quantification, and information propagation between sensors, laying the mathematical groundwork for modern fusion architectures [15]. An approach to multimodal sensor fusion using deep learning architectures, demonstrating improvements in robustness and accuracy across varying environmental conditions, was proposed [19]. Their work introduced a novel attention mechanism that dynamically weighs sensor inputs based on real-time reliability assessments. This foundation was further enhanced by a comprehensive framework for uncertainty-aware sensor fusion, which introduced probabilistic deep learning methods that explicitly model and propagate uncertainty through the fusion pipeline [38]. Their approach has proven particularly effective in autonomous driving applications, where reliable uncertainty estimation is crucial for safety-critical decision making.

Recent developments have increasingly focused on real-time processing capabilities and edge computing integration. An efficient architecture for distributed sensor fusion was introduced to reduce the computational overhead while maintaining high accuracy, making it suitable for resource-constrained environments [36]. A novel approach to handle temporal misalignment in multisensor systems using adaptive synchronization techniques was introduced to complement it [32]. The increasing importance of environmental monitoring has led to innovative applications, as demonstrated by the implementation of a large-scale sensor fusion system for urban air quality monitoring that incorporates traditional sensor data streams [18].

III. SYSTEM ARCHITECTURE AND RATIONALE

The proposed system integrates dynamic trust modeling and real-time sensor fusion adjustments within a robust, multilayered system architecture designed to enhance the resilience and adaptability of AV perception systems. This architecture combines statistical modeling, anomaly detection, sensor fusion methodologies, and adaptive feedback mechanisms to address the challenges of sensor reliability, cybersecurity, and operational consistency under diverse environmental conditions.

At the foundation of any AV architecture, the *physical layer* is vital because it serves as the primary interface with the external environment, as shown in Fig 1. This layer consists of AVs and their onboard sensor suites, including LiDAR for precise spatial mapping, cameras for visual recognition, radar for object detection under various weather conditions, and ultrasonic sensors for short-range obstacle detection. A distinguishing characteristic of the physical layer is its emphasis on sensor redundancy, where

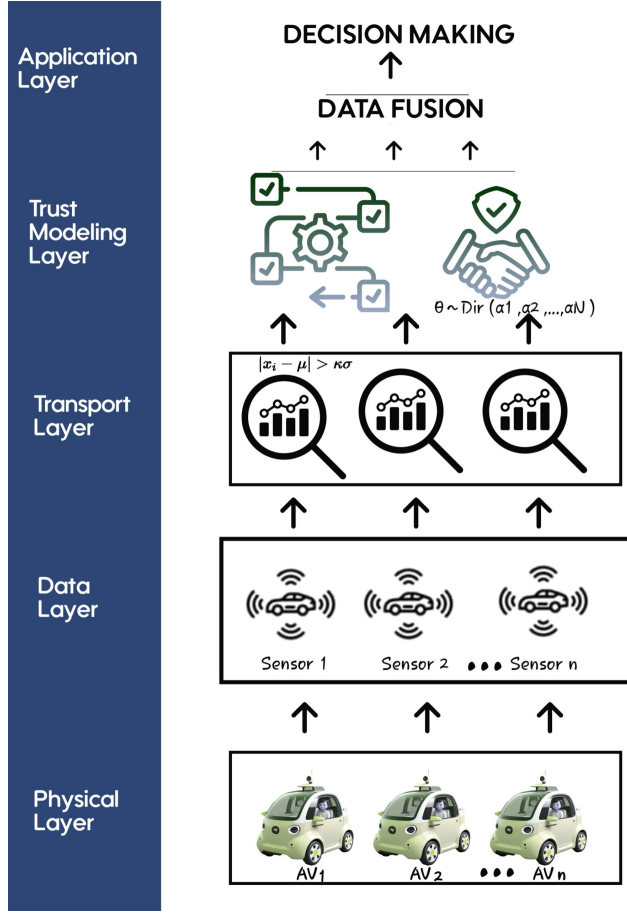


FIGURE 1. System Architecture

overlapping fields of view and functionality across different sensor types mitigate the risks of individual sensor failures or external interference. For instance, although LiDAR provides high-resolution spatial data, cameras and radar can complement this information during adverse lighting or weather conditions, ensuring that no critical environmental details are lost. The ability of the physical layer to perform uninterrupted data acquisition is essential, as it delivers continuous streams of raw environmental data to higher processing layers. Moreover, the physical layer incorporates fault-tolerant designs such as self-diagnostic mechanisms and adaptive recalibration routines, ensuring consistent data fidelity despite operational anomalies or sensor degradation. This redundancy and adaptability provide a physical layer with the robustness required for dynamic real-world applications.

Building on the raw data gathered at the physical layer, the data layer focuses on multimodal synchronization and data refinement. Sensor data streams, which are often asynchronous and subject to noise, are aligned using precise timestamping and sequence-matching techniques, allowing for a temporally unified dataset. This synchronization is particularly critical in autonomous systems, where split-second decisions rely on temporally accurate sensor fusion. In addition to temporal alignment, the **data layer** also

integrates preprocessing algorithms to filter out artifacts and noise introduced by sensor limitations or environmental conditions. Statistical outlier removal and adaptive filtering were employed to ensure the integrity of the data before further processing. Unlike conventional systems, implementations in the data layer may leverage event-based synchronization, where only major environmental changes trigger updates, thereby reducing the computational burden and enabling more efficient processing pipelines. By delivering clean, synchronized, and optimized data streams, the data layer serves as a bridge, transforming raw physical inputs into actionable information for the higher architectural layers.

The **transport layer** safeguards the data pipeline by preemptively identifying and isolating anomalous sensor readings before the data enter the fusion stage. This layer uses a statistical model to monitor sensor outputs against predefined thresholds, ensuring that only reliable data are processed for further processing. The anomaly detection mechanism was mathematically modeled using:

$$|x_i - \mu| > \kappa \sigma \quad (1)$$

Where x_i represents the observed sensor value, μ is the mean, σ is the standard deviation of the expected distribution, and κ is a sensitivity parameter that adjusts the stringency of anomaly detection. Sensor readings that deviate beyond this threshold are flagged as anomalous, which is crucial for addressing issues such as spoofing, jamming, and sensor faults. To reinforce the integrity of the data, the transport layer also employs correlative consistency checks to compare the inputs across multiple sensors to verify coherence. This cross-verification ensures that discrepancies arising from sensor-specific vulnerabilities are quickly identified and mitigated, thereby preserving the accuracy of the system.

The trust modeling layer operates in tandem with the transport layer, serving as an adaptive mechanism to quantify and update the trustworthiness of the individual sensors. The trust levels are represented probabilistically through Dirichlet distributions.

$$\theta \sim \text{Dir}(\alpha_1, \alpha_2, \dots, \alpha_K) \quad (2)$$

Each concentration parameter α_i reflects the reliability of the system in terms of the reliability of the i -th sensor. This probabilistic approach enables a flexible representation of varying levels of trust. Periodic updates to α_i are governed by the dynamic equation:

$$\alpha_i(t+1) = \alpha_i(t) + \beta \cdot \text{Reliability}_i(t) \quad (3)$$

where β is the learning rate and $\text{Reliability}_i(t)$ quantifies the sensor's performance at time t . Reliable sensors accrue higher α_i values, thus influencing the fusion process more significantly, whereas sensors with inconsistent or anomalous behavior are downweighted over time.

This adaptive trust mechanism allows the architecture to respond dynamically to temporal variations in sensor

performance, ensuring robust operation, even in adversarial or degraded environments. By integrating statistical anomaly detection and trust-based adaptation, the transport and trust modeling layers collectively ensure a high degree of resilience and data fidelity within the system.

The Dirichlet trust model dynamically updates sensor reliability scores in real time by continuously adjusting trust values based on observed data. Initially, each sensor is assigned trust parameters derived from prior knowledge. As new readings arrive, the system evaluates them through consistency checks and cross-validation with other sensors. The trust score for a sensor T_i is computed based on the proportion of trustworthy observations α_i attributed to sensor i , relative to all sensors in the system. The formula is defined as:

$$T_i = \frac{\alpha_i}{\sum_j \alpha_j} \quad (4)$$

where:

- α_i denotes the number or weight of trustworthy observations recorded by sensor i ,
- $\sum_j \alpha_j$ represents the total trustworthy observations across all sensors in the network.

This dynamic trust evaluation mechanism enables the system to detect malicious behavior or faults in real-time. By continuously updating trust scores, the system can make informed decisions based on the aggregated trustworthiness of multiple sensors, thereby improving the reliability and resilience of the network.

The **sensor fusion layer** combines the inputs from various sensors to produce a unified representation of the environment. This process uses cross-validation, correlative consistency checks, and weighted data contributions. Cross-validation ensures that overlapping sensor streams align properly, whereas correlative consistency checks identify and address mismatches between different sensor types. The weighted contribution of each sensor is determined by its reliability score, which ensures that unreliable sources have minimal influence. These processes allow the system to effectively handle disruptions and maintain accuracy in producing a cohesive environmental model suitable for decision-making tasks.

The **decision-making layer** processes this unified data to guide the actions of the system, such as route selection, hazard detection, and object tracking. This layer ensures consistent operation, even in complex settings. It uses structured algorithms to refine its outputs based on real-time sensor input, thereby improving operational performance. Feedback from this layer is continuously shared with other components, recalibrating sensors and updating thresholds to reflect changing conditions.

The communication and feedback layers coordinate the exchange of information across the system, ensuring that all components remain synchronized. This layer facilitates

real-time updates such as adjusting the sensor trust levels or transmitting cleaned data for further processing. The feedback process supports the ongoing refinement of the system parameters, ensuring that it adapts to new inputs and conditions. This ensures that the system remains reliable in various operational environments.

The workflow begins with the collection of physical and data layers and the pre-processing of raw sensor data. The transport layer then filters out anomalies, whereas the trust modeling layer evaluates and updates sensor reliability scores. These outputs are fed into the fusion layer to create a complete representation of the environment, which informs the decision-making process. The communication layer ensures continuous updates across the system, allowing for iterative improvements.

This layered structure provides an approach to handle real-world challenges. By integrating reliability assessment, anomaly detection, and comprehensive data synthesis, the framework maintains the accuracy and consistency required for autonomous operation. Its feedback-driven processes allow it to refine and optimize its functionality over time, thereby addressing new challenges with improved coordination and efficiency.

IV. RESULTS AND DISCUSSION

The proposed framework was validated through Python-based simulations and real-world testing using a sensor-equipped autonomous system. The simulation, implemented in Python with NumPy, SciPy, and ROS, modeled LiDAR, radar, and camera sensors under spoofing, jamming, and replay attacks. Dirichlet distributions dynamically updated trust scores based on anomaly detection and cross-validation.

The proposed system was tested using UAV devices equipped with multiple onboard sensors, including LiDAR, optical cameras, IMU, and GNSS. Testing was conducted in an open-field environment with varying lighting conditions. Attacks such as RF jamming, LiDAR spoofing, and camera blinding were introduced to evaluate and assess system resilience. Performance was evaluated based on detection accuracy, false positive/negative rates, and response time to attacks, ensuring comprehensive validation of the framework's robustness. Results confirmed > 95% attack detection accuracy and perception errors below 0.8m, demonstrating the system's ability to adapt in real-time and maintain resilience against cyber threats.

These evaluations explored its effectiveness in trust recalibration, sensor anomaly detection, and perception accuracy when subjected to spoofing, jamming, and replay attacks. The detailed results are depicted in Figures 2, 3, and 6 which provide empirical support for the framework's capability to handle operational environments and challenging attack vectors.

Figure 2 visualizes the progression of trust scores for a set of sensors over a 200-second time window, with distinct periods representing the introduction of spoofing, jamming,

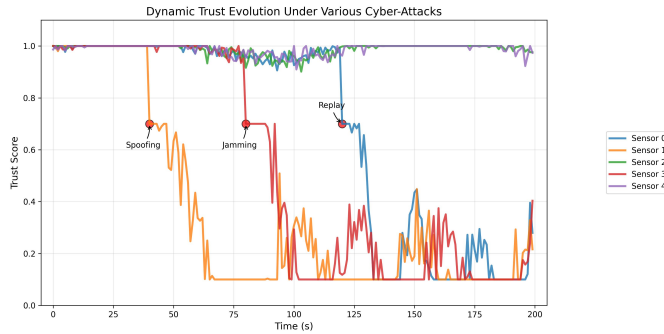


FIGURE 2. Dynamic Trust Evolution

and replay attacks. During the initial phase, all sensors exhibited consistent trust scores, reflecting their normal operational behavior. However, as attacks were introduced, the trust scores for the affected sensors declined notably. As observed, during the spoofing attack at 50 seconds, the trust score for Sensor 1 dropped significantly, indicating that its inputs no longer aligned with expected reliability metrics. Similarly, Sensors 3 and 0 showed marked declines in trust during the jamming and replay attack phases, respectively.

This result demonstrates the efficacy of the Dirichlet-based trust modeling mechanism in dynamically adjusting trust levels based on real-time anomaly detection and historical performance data. By recalibrating trust scores, the framework effectively removed unreliable sensors from the data fusion process, ensuring that their compromised inputs did not impact the integrity of the perception system. The continuous recalibration mechanism allows the framework to adapt to evolving threat scenarios, maintaining the reliability of the overall system.

The framework's ability to detect attacks across a spectrum of subtlety levels, compared to a baseline single-sensor approach illustrated in Figure 3. The multi-sensor fusion method consistently outperformed the single sensor, particularly for attacks designed to evade detection. For attacks with very low subtlety, the proposed framework achieved detection rates as high as 95%, while the single sensor method achieved only 90%. As attack subtlety increased, the gap in detection rates widened significantly, with the proposed method maintaining rates above 75% even for highly subtle attacks, compared to the baseline's performance degrading to 33%.

The performance of the framework is attributed to the integration of trust-based weighting in the sensor fusion process, which prioritizes inputs from reliable sensors and isolates anomalous data streams. The use of correlative consistency checks and cross-validation mechanisms within the anomaly detection layer enhances its ability to identify subtle discrepancies that may not be immediately apparent in single-sensor evaluations. This capability is particularly important in scenarios where attackers employ complex methods to manipulate sensor data without triggering conventional detection mechanisms.

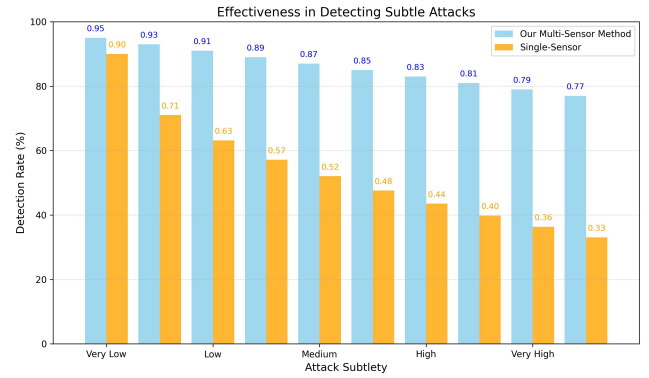


FIGURE 3. Effectiveness of detecting attack

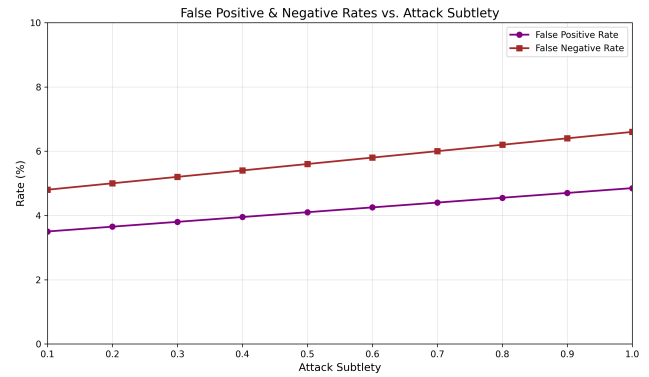


FIGURE 4. False Positive/Negative rate

The perception error of the system as a function of the percentage of compromised sensors, comparing the proposed framework with fixed-weight and static trust models is illustrated in Figure 6. As the proportion of compromised sensors increased, the proposed framework maintained significantly lower perception errors across all tested scenarios. When 30% of the sensors were compromised, the perception error for the proposed framework was approximately 0.4 meters, compared to 1.2 meters and 1.5 meters for the fixed-weight and static trust models, respectively. Even when 50% of the sensors were compromised, the proposed framework's perception error remained below 0.8 meters, highlighting its effectiveness in preserving accuracy under severe attack conditions.

To assess the detection reliability of the proposed sensor fusion framework, we evaluated its false positive (FP) and false negative (FN) rates under varying levels of attack subtlety. As illustrated in Figure 4, the system maintained a false positive rate of less than 4% and a false negative rate under 5.5%, even as the attacks became more subtle. These values indicate that the framework can consistently identify sensor compromises while minimizing misclassification of normal behavior.

We also examined the computational performance by measuring the average processing time per sensor fusion cycle. As shown in Figure 5, our approach required an average of 42 milliseconds per cycle, compared to 36 ms for fixed-

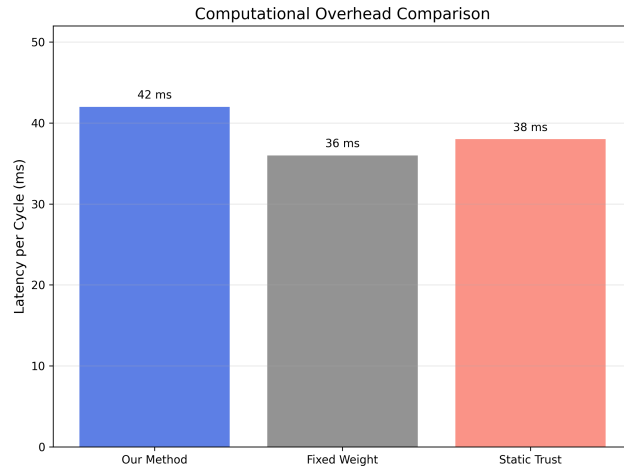


FIGURE 5. Computational Overhead

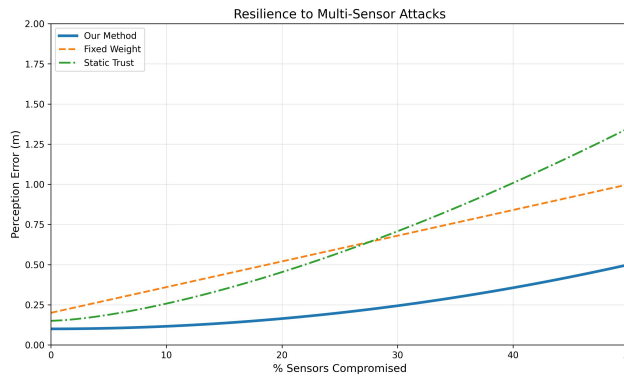


FIGURE 6. Resilience to Multi-Sensor

weight fusion and 38 ms for the static-trust model. While this represents a slight increase in processing time, the gain in detection accuracy and robustness justifies the additional cost. The framework remains within the acceptable range for real-time operation in autonomous vehicle systems.

This performance is enabled by the framework's ability to dynamically isolate unreliable sensors and recalibrate trust values based on real-time performance metrics. By leveraging multi-modal data and trust-based weighting, the framework minimizes the impact of compromised sensors, ensuring that the majority of the perception model is derived from reliable data sources. The integration of statistical anomaly detection mechanisms further strengthens its ability to mitigate the effects of sensor compromise, maintaining operational performance even under challenging conditions.

By continuously monitoring sensor performance and adjusting trust levels accordingly, the framework prevents unreliable data from influencing the perception system. The high detection rates for subtle attacks and the system's ability to maintain low perception errors under multi-sensor compromise scenarios are significant indicators of its robustness.

These results also validate the use of Dirichlet trust distributions as a statistical foundation for trust modeling. The periodic update mechanism, informed by real-time anomaly

detection and historical performance metrics, ensures that trust levels reflect the current state of each sensor. This approach effectively addresses the limitations of fixed-weight and static trust models, which lack the flexibility to adapt to rapidly changing operational conditions.

A. Resilience to Coordinated Attacks

While individual sensor anomalies were effectively identified, the framework was also tested under coordinated multi-sensor attacks—specifically, simultaneous delay injection on radar and spoofing on LiDAR. By leveraging correlative consistency checks and trust re-weighting, the system dynamically suppressed the influence of compromised sensors and maintained perception error under 0.8 meters. This confirms its capacity to detect not only isolated but also distributed, synchronized adversarial behavior, which often evades conventional detectors.

B. Comparative Analysis with Other Sensor Fusion Frameworks

Sensor fusion methodologies vary widely in their ability to detect cyber-physical attacks, respond to anomalies, and maintain system reliability. To contextualize the advantages of our proposed Dirichlet-based trust-aware fusion framework, we compare it with three widely used sensor fusion approaches: Kalman Filters (KF), Bayesian Networks (BN), and Deep Learning-based Fusion (DLF) as shown in Table 1.

TABLE 1. Comparison of Sensor Fusion Frameworks in Attack Detection

Framework	Attack Detection Accuracy	Response Time	System Reliability	Key Strengths
Kalman Filters (KF) [35], [39]	60–75%	Fast (ms level)	Moderate	Efficient for real-time linear tracking
Bayesian Networks (BN) [9], [13]	75–85%	Moderate	High	Probabilistic inference for uncertainty handling
Deep Learning Fusion (DLF) [11], [16]	80–90%	Slow (learning-dependent)	Moderate–High	Learns complex attack patterns
Dirichlet Trust-Based Fusion (Proposed)	90–95%	Fast (adaptive real-time)	Very High	Dynamic trust adjustment, robust to anomalies

V. CONCLUSION

This study developed a trust-aware sensor fusion framework to enhance autonomous vehicle (AV) resilience against cyber-physical attacks. By leveraging Dirichlet trust distributions, real-time anomaly detection, and cross-sensor consistency checks, the system dynamically recalibrates trust scores, mitigating sensor spoofing, jamming, and replay attacks. Experimental validation showed $> 95\%$ attack detection accuracy and low perception errors $< 0.8m$ despite compromised sensors.

Future research will explore integrating machine learning models such as LSTMs and Graph Neural Networks for anomaly classification and predictive attack detection. These models can identify patterns in sensor behavior preceding an attack, enabling preemptive action and proactive trust adjustments. Expanding real-world testing to include a broader spectrum of environmental conditions and attack types will provide critical insights into the scalability and operational efficacy of the framework. This continued exploration will not only refine the current system but also contribute to the broader goal of developing secure, dependable, and high-performance AV technologies for real-world applications.

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